mechanism, also referred to as its mobility, is treated in most texts on mechanism analysis and design, e.g., [43, 114]. The notion of a robot's configuration space was first formulated in the context of motion planning by Lozano-Perez [95]; more recent and advanced treatments can be found in [80, 83, 27]. As apparent from some of the examples in this chapter, a robot's configuration space can be nonlinear and curved, as can its task space. Such spaces often have the mathematical structure of a differentiable manifold, which are the central objects of study in differential geometry. Some accessible introductions to differential geometry are [119, 38, 17].

2.8 Exercises

In the exercises below, if you are asked to "describe" a C-space, you should indicate its dimension and whatever you know about its topology (e.g., using $\mathbb{R}$, S, and T, as with the examples in Sections 2.3.1 and 2.3.2).

Exercise 2.1 Using the methods of Section 2.1 derive a formula, in terms of n, for the number of degrees of freedom of a rigid body in n-dimensional space. Indicate how many of these dof are translational and how many are rotational. Describe the topology of the C-space (e.g., for $n = 2$, the topology is $\mathbb{R}^2 \times S^1$).

Exercise 2.2 Find the number of degrees of freedom of your arm, from your torso to your palm (just past the wrist, not including finger degrees of freedom). Keep the center of the ball-and-socket joint of your shoulder stationary (do not "hunch" your shoulders). Find the number of degrees of freedom in two ways:

(a) add up the degrees of freedom at the shoulder, elbow, and wrist joints;

(b) fix your palm flat on a table with your elbow bent and, without moving the center of your shoulder joint, investigate with how many degrees of freedom you can still move your arm.

Do your answers agree? How many constraints were placed on your arm when you placed your palm at a fixed configuration on the table?

Exercise 2.3 In the previous exercise, we assumed that your arm is a serial chain. In fact, between your upper arm bone (the humerus) and the bone complex just past your wrist (the carpal bones), your forearm has two bones, the radius and the ulna, which are part of a closed chain. Model your arm, from your shoulder to your palm, as a mechanism with joints and calculate the number of degrees of freedom using Grübler's formula. Be clear on the number of freedoms of each joint you use in your model. Your joints may or may not be

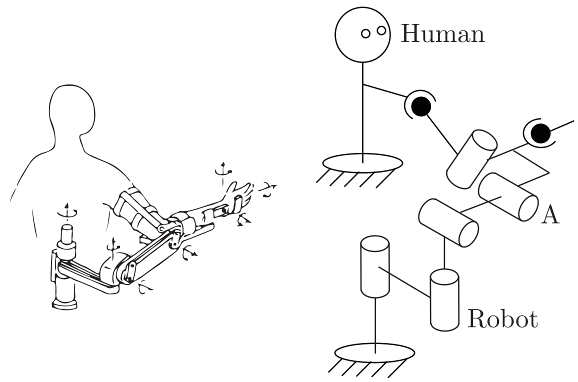

Figure 2.15: Robot used for human arm rehabilitation.

of the standard types studied in this chapter (R, P, H, C, U, and S).

Exercise 2.4 Assume each of your arms has n degrees of freedom. You are driving a car, your torso is stationary relative to the car (owing to a tight seatbelt!), and both hands are firmly grasping the wheel, so that your hands do not move relative to the wheel. How many degrees of freedom does your arms-plus-steering wheel system have? Explain your answer.

Exercise 2.5 Figure 2.15 shows a robot used for human arm rehabilitation. Determine the number of degrees of freedom of the chain formed by the human arm and the robot

Exercise 2.6 The mobile manipulator of Figure 2.16 consists of a 6R arm and multi-fingered hand mounted on a mobile base with a single wheel. You can think of the wheeled base as the same as the rolling coin in Figure 2.11 – the wheel (and base) can spin together about an axis perpendicular to the ground, and the wheel rolls without slipping. The base always remains horizontal. (Left unstated are the means to keep the base horizontal and to spin the wheel and base about an axis perpendicular to the ground.)

(a) Ignoring the multi-fingered hand, describe the configuration space of the mobile manipulator.

(b) Now suppose that the robot hand rigidly grasps a refrigerator door handle and, with its wheel and base completely stationary, opens the door using only its arm. With the door open, how many degrees of freedom does the

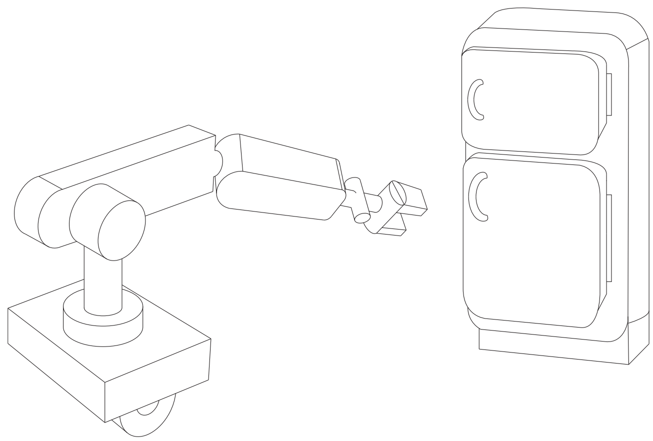

Figure 2.16: Mobile manipulator.

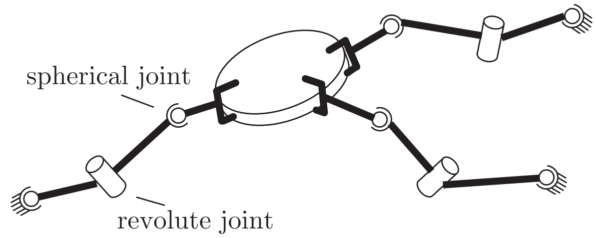

Figure 2.17: Three cooperating SRS arms grasping a common object.

mechanism formed by the arm and open door have?

(c) A second identical mobile manipulator comes along, and after parking its mobile base, also rigidly grasps the refrigerator door handle. How many degrees of freedom does the mechanism formed by the two arms and the open refrigerator door have?

Exercise 2.7 Three identical SRS open-chain arms are grasping a common object, as shown in Figure 2.17.

(a) Find the number of degrees of freedom of this system.
(b) Suppose there are now a total of n such arms grasping the object. How many degrees of freedom does this system have?
(c) Suppose the spherical wrist joint in each of the n arms is now replaced by

a universal joint. How many degrees of freedom does this system have?

Exercise 2.8 Consider a spatial parallel mechanism consisting of a moving plate connected to a fixed plate by n identical legs. For the moving plate to have six degrees of freedom, how many degrees of freedom should each leg have, as a function of n? For example, if $n = 3$ then the moving plate and fixed plate are connected by three legs; how many degrees of freedom should each leg have for the moving plate to move with six degrees of freedom? Solve for arbitrary n.

Exercise 2.9 Use the planar version of Grübler's formula to determine the number of degrees of freedom of the mechanisms shown in Figure 2.18. Comment on whether your results agree with your intuition about the possible motions of these mechanisms.

Exercise 2.10 Use the planar version of Grübler's formula to determine the number of degrees of freedom of the mechanisms shown in Figure 2.19. Comment on whether your results agree with your intuition about the possible motions of these mechanisms.

Exercise 2.11 Use the spatial version of Grübler's formula to determine the number of degrees of freedom of the mechanisms shown in Figure 2.20. Comment on whether your results agree with your intuition about the possible motions of these mechanisms.

Exercise 2.12 Use the spatial version of Grübler's formula to determine the number of degrees of freedom of the mechanisms shown in Figure 2.21. Comment on whether your results agree with your intuition about the possible motions of these mechanisms.

Exercise 2.13 In the parallel mechanism shown in Figure 2.22, six legs of identical length are connected to a fixed and moving platform via spherical joints. Determine the number of degrees of freedom of this mechanism using Grübler's formula. Illustrate all possible motions of the upper platform.

Exercise 2.14 The $3 \times$UPU platform of Figure 2.23 consists of two platforms–the lower one stationary, the upper one mobile–connected by three UPU legs.

(a) Using the spatial version of Grübler's formula, verify that it has three degrees of freedom.

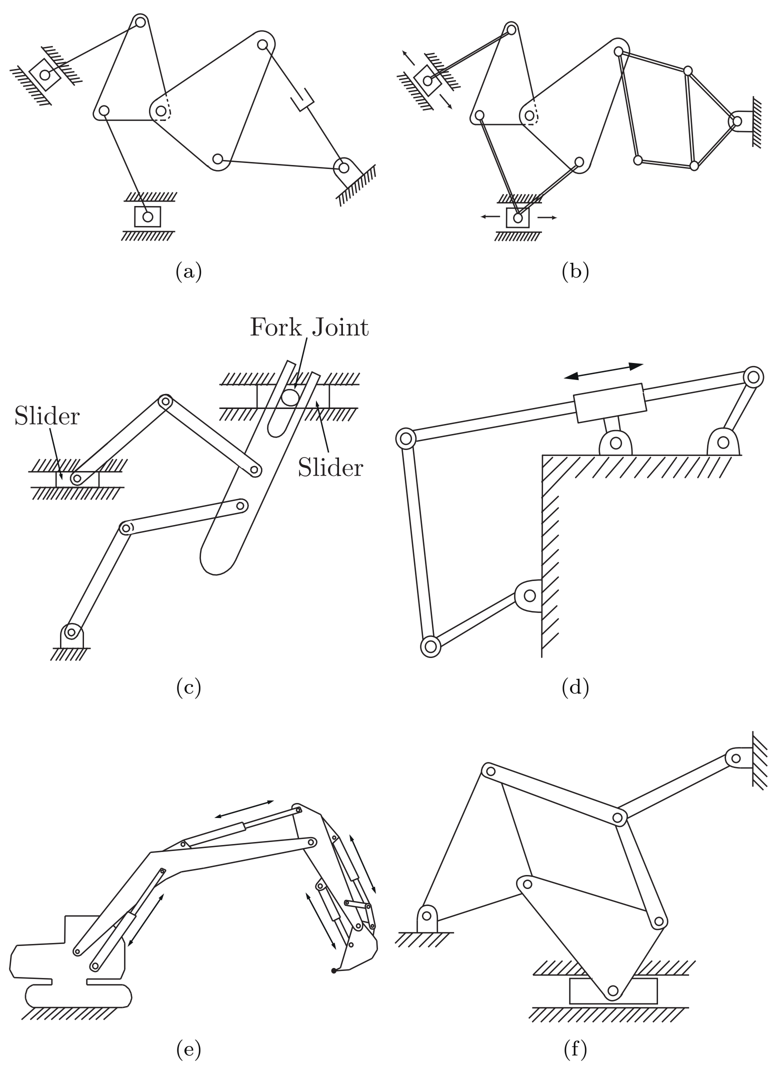

Figure 2.18: A first collection of planar mechanisms.

(b) Construct a physical model of the $3 \times$ UPU platform to see if it indeed has three degrees of freedom. In particular, lock the three P joints in place; does the robot become a rigid structure as predicted by Grübler's formula,

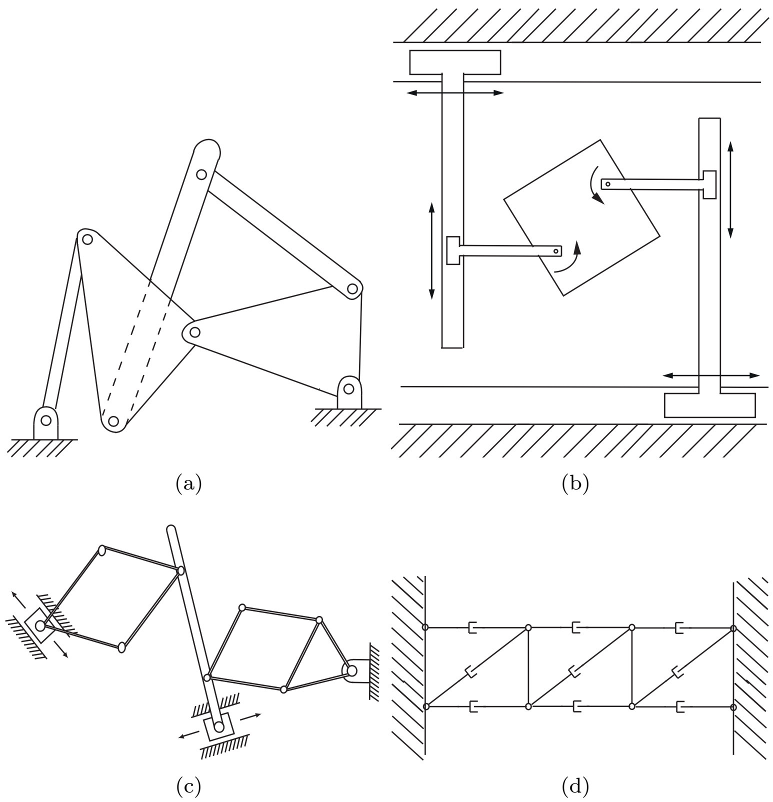

Figure 2.19: A second collection of planar mechanisms.

or does it move?

Exercise 2.15 The mechanisms of Figures 2.24(a) and 2.24(b).

(a) The mechanism of Figure 2.24(a) consists of six identical squares arranged in a single closed loop, connected by revolute joints. The bottom square is fixed to ground. Determine the number of degrees of freedom using Grübler's formula.

(b) The mechanism of Figure 2.24(b) also consists of six identical squares connected by revolute joints, but arranged differently (as shown). Determine the number of degrees of freedom using Grübler's formula. Does your result agree with your intuition about the possible motions of this

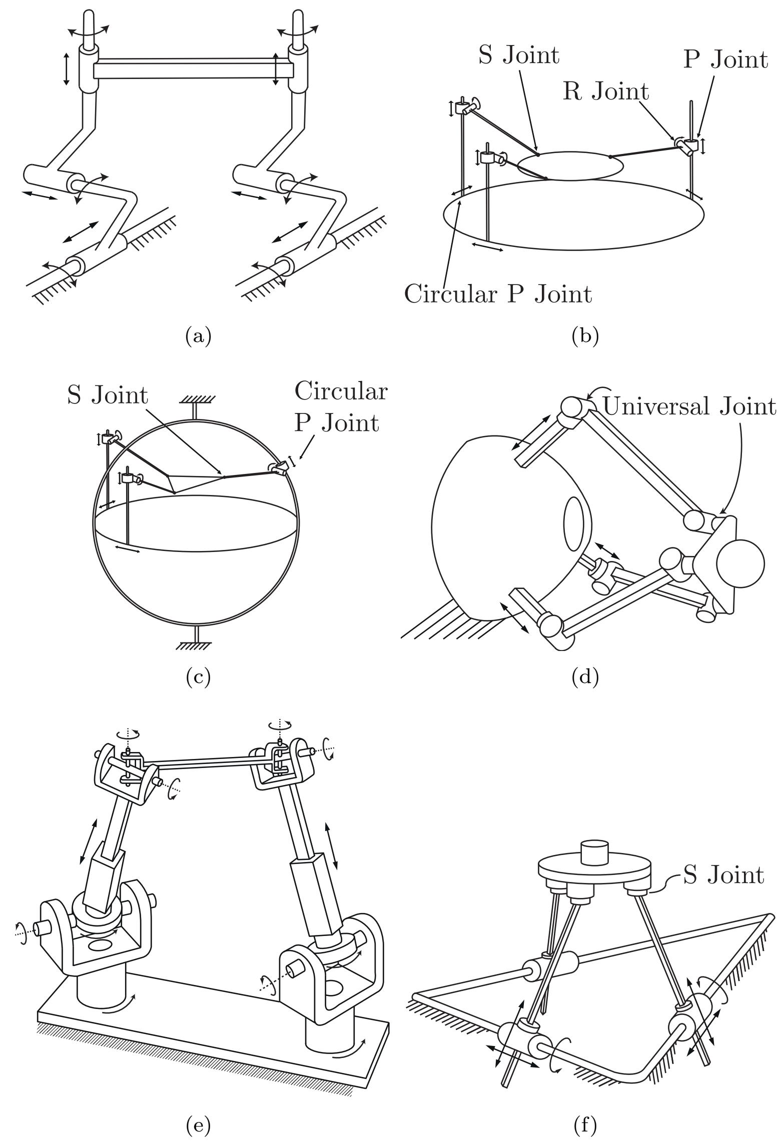

Figure 2.20: A first collection of spatial parallel mechanisms.

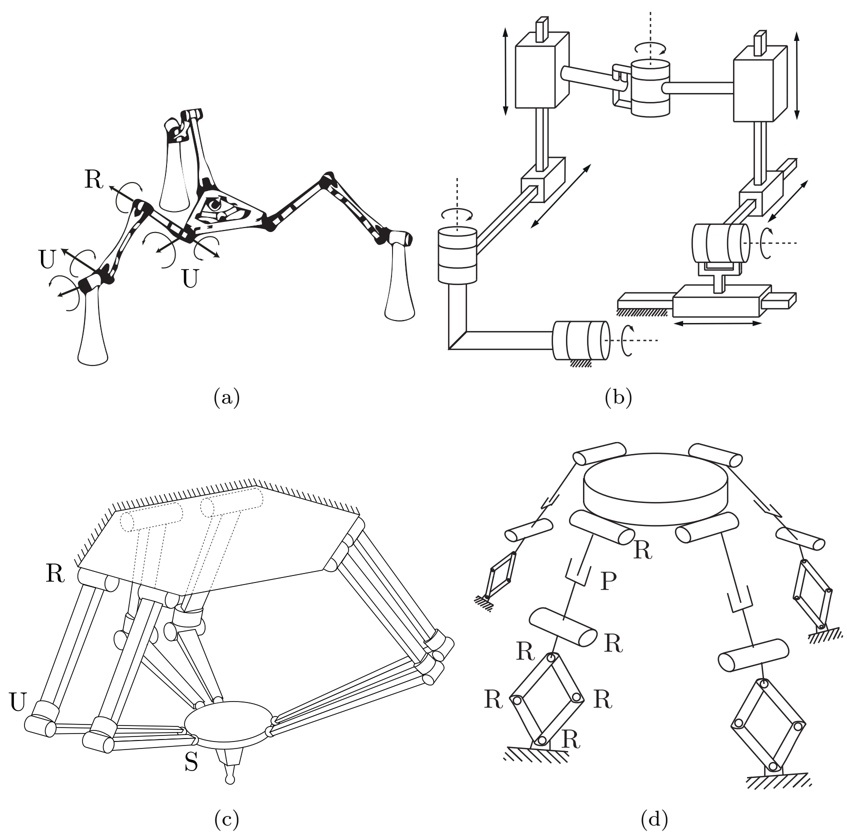

Figure 2.21: A second collection of spatial parallel mechanisms.

mechanism?

Exercise 2.16 Figure 2.25 shows a spherical four-bar linkage, in which four links (one of the links is the ground link) are connected by four revolute joints to form a single-loop closed chain. The four revolute joints are arranged so that they lie on a sphere such that their joint axes intersect at a common point.

(a) Use Grübler's formula to find the number of degrees of freedom. Justify your choice of formula.

(b) Describe the configuration space.

(c) Assuming that a reference frame is attached to the center link, describe

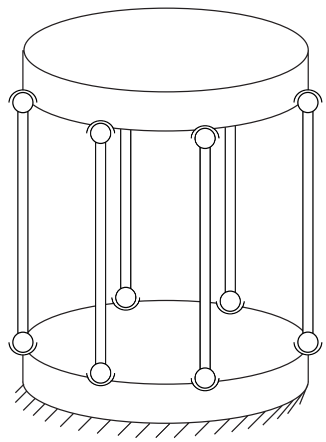

Figure 2.22: A 6×SS platform.

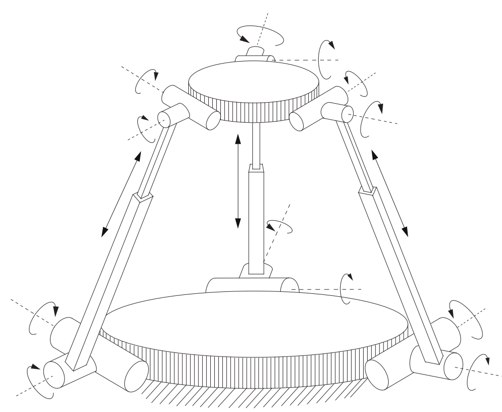

Figure 2.23: The 3×UPU platform.

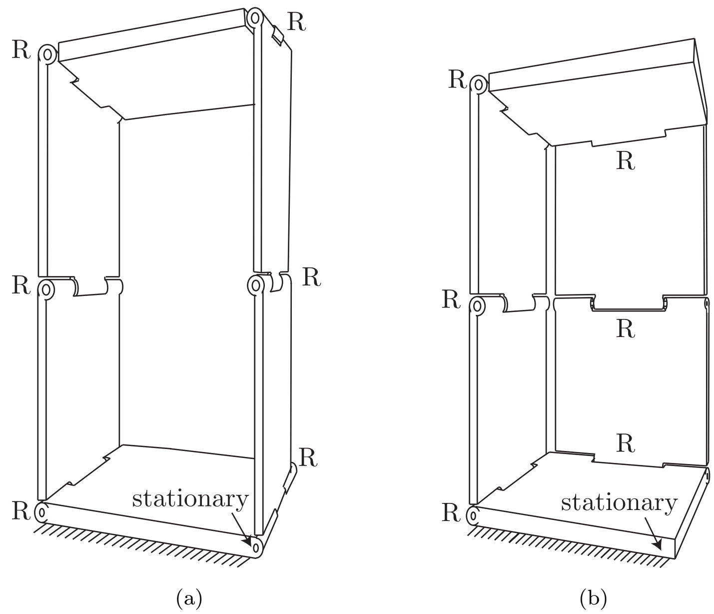

Figure 2.24: Two mechanisms.

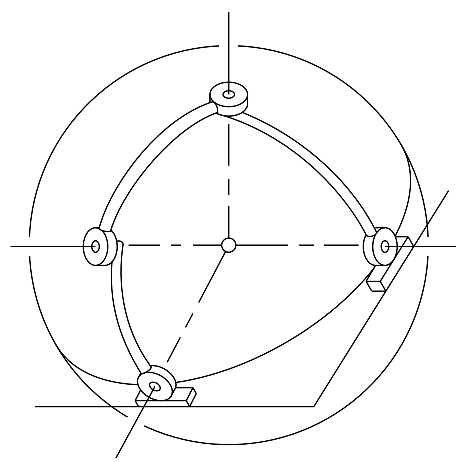

Figure 2.25: The spherical four-bar linkage.

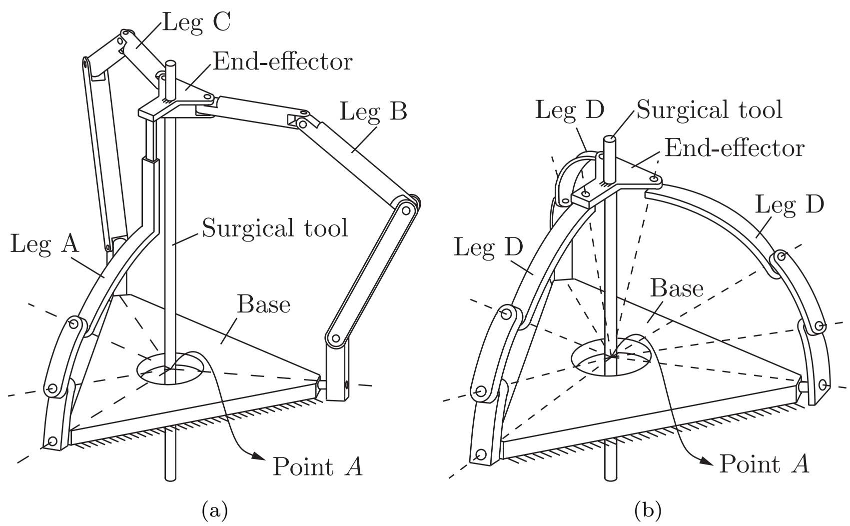

Figure 2.26: Surgical manipulator.

its workspace.

Exercise 2.17 Figure 2.26 shows a parallel robot used for surgical applications. As shown in Figure 2.26(a), leg A is an RRRP chain, while legs B and C are RRRUR chains. A surgical tool is rigidly attached to the end-effector.

(a) Use Grübler's formula to find the number of degrees of freedom of the mechanism in Figure 2.26(a).

(b) Now suppose that the surgical tool must always pass through point A in Figure 2.26(a). How many degrees of freedom does the manipulator have?

(c) Legs A, B, and C are now replaced by three identical RRRR legs as shown in Figure 2.26(b). Furthermore, the axes of all R joints pass through point A. Use Grübler's formula to derive the number of degrees of freedom of this mechanism.

Exercise 2.18 Figure 2.27 shows a $3 \times$ PUP platform, in which three identical PUP legs connect a fixed base to a moving platform. The P joints on both the fixed base and moving platform are arranged symmetrically. Use Grübler's formula to find the number of degrees of freedom. Does your answer agree with your intuition about this mechanism? If not, try to explain any discrepancies without resorting to a detailed kinematic analysis.

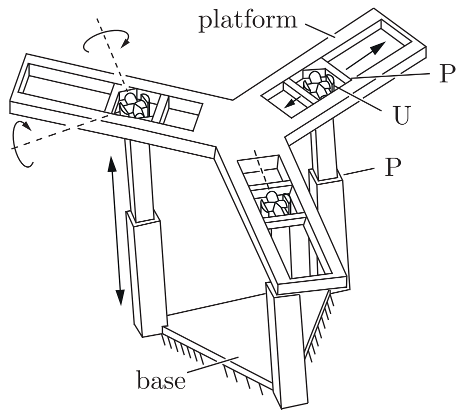

Figure 2.27: The 3×PUP platform.

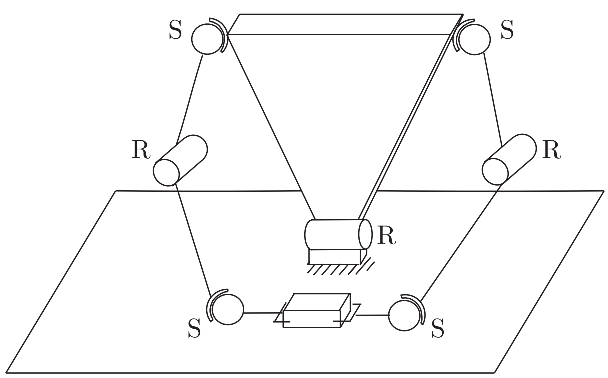

Figure 2.28: Dual-arm robot.

Exercise 2.19 The dual-arm robot of Figure 2.28 is rigidly grasping a box. The box can only slide on the table; the bottom face of the box must always be in contact with the table. How many degrees of freedom does this system have?

Exercise 2.20 The dragonfly robot of Figure 2.29 has a body, four legs, and four wings as shown. Each leg is connected to each adjacent leg by a USP linkage. Use Grübler's formula to answer the following questions.

(a) Suppose the body is fixed and only the legs and wings can move. How

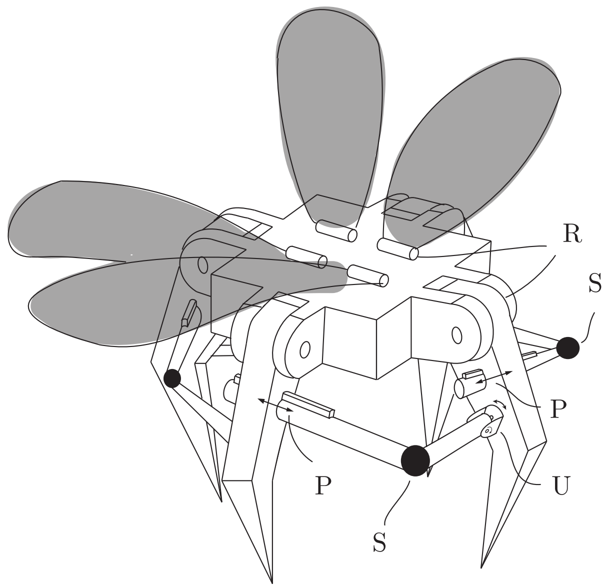

Figure 2.29: Dragonfly robot.

many degrees of freedom does the robot have?

(b) Now suppose the robot is flying in the air. How many degrees of freedom does the robot have?

(c) Now suppose the robot is standing with all four feet in contact with the ground. Assume that the ground is uneven and that each foot–ground contact can be modeled as a point contact with no slip. How many degrees of freedom does the robot have? Explain your answer.

Exercise 2.21 A caterpillar robot.

(a) A caterpillar robot is hanging by its tail end as shown in Figure 2.30(a). The robot consists of eight serially connected rigid links (one head, one tail, and six body links). The six body links are connected by revolute–prismatic–revolute joints, while the head and tail are connected to the body by revolute joints. Find the number of degrees of freedom of this robot.

(b) The caterpillar robot is now crawling on a leaf as shown in Figure 2.30(b). Suppose that all six body links must make contact with the leaf at all times but the links can slide and rotate on the leaf. Find the number of

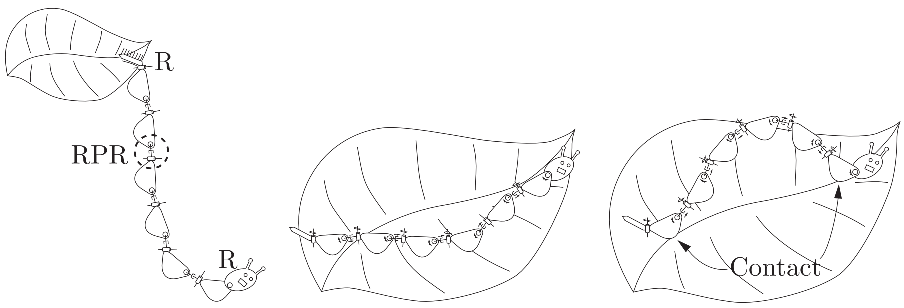

Figure 2.30: A caterpillar robot.

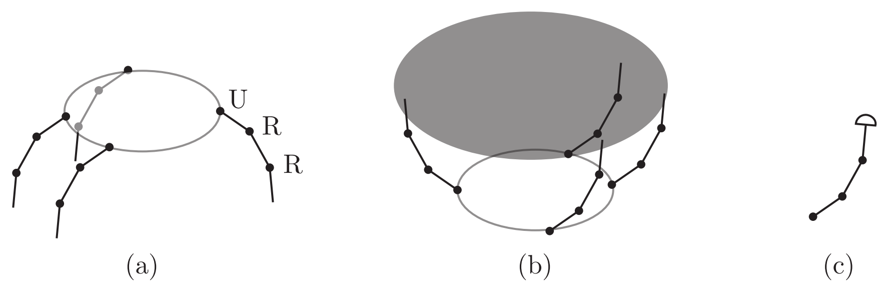

Figure 2.31: (a) A four-fingered hand with palm. (b) The hand grasping an ellipsoidal object. (c) A rounded fingertip that can roll on the object without sliding.

degrees of freedom of this robot during crawling.

(c) Now suppose the caterpillar robot crawls on the leaf as shown in Figure 2.30(c), with only the first and last body links in contact with the leaf. Find the number of degrees of freedom of this robot during crawling.

Exercise 2.22 The four-fingered hand of Figure 2.31(a) consists of a palm and four URR fingers (the U joints connect the fingers to the palm).

(a) Assume that the fingertips are points and that one fingertip is in contact with the table surface (sliding of the fingertip point-contact is allowed). How many degrees of freedom does the hand have? What if two fingertips are in sliding point contact with the table? Three? All four?

(b) Repeat part (a) but with each URR finger replaced by an SRR finger (each universal joint is replaced by a spherical joint).

(c) The hand (with URR fingers) now grasps an ellipsoidal object, as shown in Figure 2.31(b). Assume that the palm is fixed in space and that no slip occurs between the fingertips and object. How many degrees of freedom does the system have?

(d) Now assume that the fingertips are hemispheres as shown in Figure 2.31(c). Each fingertip can roll on the object but cannot slip or break contact with the object. How many degrees of freedom does the system have? For a single fingertip in rolling contact with the object, comment on the dimension of the space of feasible fingertip velocities relative to the object versus the number of parameters needed to represent the fingertip configuration relative to the object (the number of degrees of freedom). (Hint: You may want to experiment by rolling a ball around on a tabletop to get some intuition.)

Exercise 2.23 Consider the slider–crank mechanism of Figure 2.4(b). A rotational motion at the revolute joint fixed to ground (the "crank") causes a translational motion at the prismatic joint (the "slider"). Suppose that the two links connected to the crank and slider are of equal length. Determine the configuration space of this mechanism, and draw its projected version on the space defined by the crank and slider joint variables.

Exercise 2.24 The planar four-bar linkage.

(a) Use Grübler's formula to determine the number of degrees of freedom of a planar four-bar linkage floating in space.

(b) Derive an implicit parametrization of the four-bar's configuration space as follows. First, label the four links 1, 2, 3, and 4, and choose three points A, B, C on link 1, D, E, F on link 2, G, H, I on link 3, and J, K, L on link 4. The four-bar linkage is constructed in such a way that the following four pairs of points are each connected by a revolute joint: C with D, F with G, I with J, and L with A. Write down explicit constraints on the coordinates for the eight points $A, \ldots, H$ (assume that a fixed reference frame has been chosen, and denote the coordinates for point A by $p_A = (x_A, y_A, z_A)$, and similarly for the other points). Based on counting the number of variables and constraints, how many degrees of freedom does the configuration space have? If it differs from the result you obtained in (a), try to explain why.

Exercise 2.25 In this exercise we examine in more detail the representation of the C-space for the planar four-bar linkage of Figure 2.32. Attach a fixed

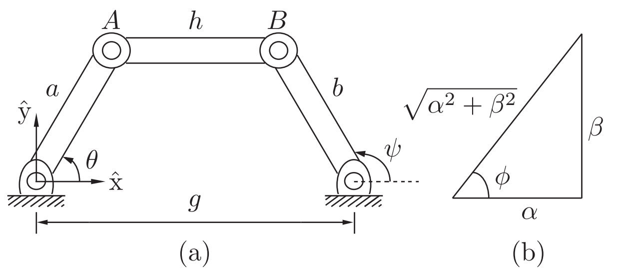

Figure 2.32: Planar four-bar linkage.

reference frame and label the joints and link lengths as shown in the figure. The (x, y) coordinates for joints A and B are given by

$$\begin{aligned} A(\theta) &= (a\cos\theta,\ a\sin\theta), \\ B(\psi) &= (g + b\cos\psi,\ b\sin\psi). \end{aligned}$$

Using the fact that the link connecting A and B is of fixed length h, i.e., $\|A(\theta) - B(\psi)\|^2 = h^2$, we have the constraint

$$b^2 + g^2 + 2gb\cos\psi + a^2 - 2(a\cos\theta(g + b\cos\psi) + ab\sin\theta\sin\psi) = h^2.$$

Grouping the coefficients of $\cos\psi$ and $\sin\psi$, the above equation can be expressed in the form

$$\alpha(\theta)\cos\psi + \beta(\theta)\sin\psi = \gamma(\theta), \tag{2.11}$$

where

$$\begin{aligned} \alpha(\theta) &= 2gb - 2ab\cos\theta, \\ \beta(\theta) &= -2ab\sin\theta, \\ \gamma(\theta) &= h^2 - g^2 - b^2 - a^2 + 2ag\cos\theta. \end{aligned}$$

We now express ψ as a function of θ, by first dividing both sides of Equation (2.11) by $\sqrt{\alpha^2 + \beta^2}$ to obtain

$$\frac{\alpha}{\sqrt{\alpha^2 + \beta^2}}\cos\psi + \frac{\beta}{\sqrt{\alpha^2 + \beta^2}}\sin\psi = \frac{\gamma}{\sqrt{\alpha^2 + \beta^2}}. \tag{2.12}$$

Referring to Figure 2.32(b), the angle ϕ is given by $\phi = \tan^{-1}(\beta/\alpha)$, so that Equation (2.12) becomes

$$\cos(\psi - \phi) = \frac{\gamma}{\sqrt{\alpha^2 + \beta^2}}.$$

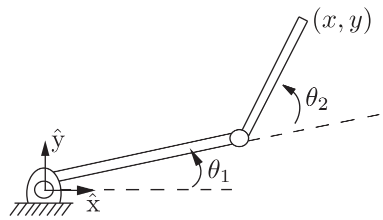

Figure 2.33: Two-link planar 2R open chain.

Therefore

$$\psi = \tan^{-1}\left(\frac{\beta}{\alpha}\right) \pm \cos^{-1}\left(\frac{\gamma}{\sqrt{\alpha^2+\beta^2}}\right).$$

(a) Note that a solution exists only if $\gamma^2 \leq \alpha^2 + \beta^2$. What are the physical implications if this constraint is not satisfied?
(b) Note that, for each value of the input angle θ, there exist two possible values of the output angle ψ. What do these two solutions look like?
(c) Draw the configuration space of the mechanism in θ–ψ space for the following link length values: $a = b = g = h = 1$.
(d) Repeat (c) for the following link length values: $a = 1$, $b = 2$, $h = \sqrt{5}$, $g = 2$.
(e) Repeat (c) for the following link length values: $a = 1$, $b = 1$, $h = 1$, $g = \sqrt{3}$.

Exercise 2.26 The tip coordinates for the two-link planar 2R robot of Figure 2.33 are given by

$$\begin{aligned} x &= 2\cos\theta_1 + \cos(\theta_1+\theta_2) \\ y &= 2\sin\theta_1 + \sin(\theta_1+\theta_2). \end{aligned}$$

(a) What is the robot's configuration space?
(b) What is the robot's workspace (i.e., the set of all points reachable by the tip)?
(c) Suppose infinitely long vertical barriers are placed at $x = 1$ and $x = -1$. What is the free C-space of the robot (i.e., the portion of the C-space that does not result in any collisions with the vertical barriers)?

Exercise 2.27 The workspace of a planar 3R open chain.

(a) Consider a planar 3R open chain with link lengths (starting from the fixed base joint) 5, 2, and 1, respectively. Considering only the Cartesian point of the tip, draw its workspace.
(b) Now consider a planar 3R open chain with link lengths (starting from the fixed base joint) 1, 2, and 5, respectively. Considering only the Cartesian point of the tip, draw its workspace. Which of these two chains has a larger workspace?
(c) A not-so-clever designer claims that he can make the workspace of any planar open chain larger simply by increasing the length of the last link. Explain the fallacy behind this claim.

Exercise 2.28 Task space.
(a) Describe the task space for a robot arm writing on a blackboard.
(b) Describe the task space for a robot arm twirling a baton.

Exercise 2.29 Give a mathematical description of the topologies of the C-spaces of the following systems. Use cross products, as appropriate, of spaces such as a closed interval $[a, b]$ of a line and $\mathbb{R}^k$, S^m, and T^n, where k, m, and n are chosen appropriately.
(a) The chassis of a car-like mobile robot rolling on an infinite plane.
(b) The car-like mobile robot (chassis only) driving around on a spherical asteroid.
(c) The car-like mobile robot (chassis only) on an infinite plane with an RRPR robot arm mounted on it. The prismatic joint has joint limits, but the revolute joints do not.
(d) A free-flying spacecraft with a 6R arm mounted on it and no joint limits.

Exercise 2.30 Describe an algorithm that drives the rolling coin of Figure 2.11 from any arbitrary initial configuration in its four-dimensional C-space to any arbitrary goal configuration, despite the two nonholonomic constraints. The control inputs are the rolling speed $\dot{\theta}$ and the turning speed $\dot{\phi}$. You should explain clearly in words or pseudocode how the algorithm would work. It is not necessary to give actual code or formulas.

Exercise 2.31 A differential-drive mobile robot has two wheels that do not steer but whose speeds can be controlled independently. The robot goes forward and backward by spinning the wheels in the same direction at the same speed, and it turns by spinning the wheels at different speeds. The configuration of the robot is given by five variables: the (x, y) location of the point halfway between

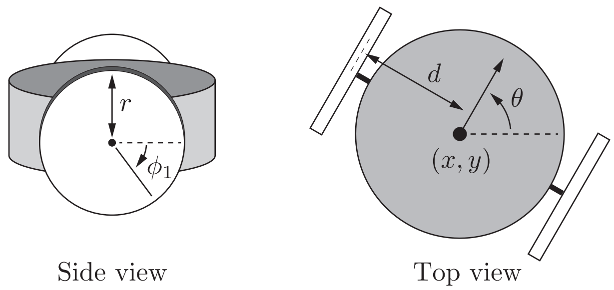

Figure 2.34: A side view and a top view of a diff-drive robot.

the wheels, the heading direction θ of the robot's chassis relative to the x-axis of the world frame, and the rotation angles ϕ_1 and ϕ_2 of the two wheels about the axis through the centers of the wheels (Figure 2.34). Assume that the radius of each wheel is r and the distance between the wheels is $2d$.

(a) Let $q = (x, y, \theta, \phi_1, \phi_2)$ be the configuration of the robot. If the two control inputs are the angular velocities of the wheels $\omega_1 = \dot{\phi}_1$ and $\omega_2 = \dot{\phi}_2$, write down the vector differential equation $\dot{q} = g_1(q)\omega_1 + g_2(q)\omega_2$. The vector fields $g_1(q)$ and $g_2(q)$ are called *control vector fields* (see Section 13.3) and express how the system moves when the respective unit control signal is applied.

(b) Write the corresponding Pfaffian constraints $A(q)\dot{q} = 0$ for this system. How many Pfaffian constraints are there?

(c) Are the constraints holonomic or nonholonomic? Or how many are holonomic and how many nonholonomic?

Exercise 2.32 Determine whether the following differential constraints are holonomic or nonholonomic:

(a)
$$(1 + \cos q_1)\dot{q}_1 + (1 + \cos q_2)\dot{q}_2 + (\cos q_1 + \cos q_2 + 4)\dot{q}_3 = 0.$$

(b)
$$\begin{aligned} -\dot{q}_1 \cos q_2 + \dot{q}_3 \sin(q_1 + q_2) - \dot{q}_4 \cos(q_1 + q_2) &= 0 \\ \dot{q}_3 \sin q_1 - \dot{q}_4 \cos q_1 &= 0. \end{aligned}$$